

20. (b) 21. (d) 22. (d) 23. (b) 24. (b) 25. (b) 26. (a) 27. (a) 28. (a) 29. (c)
 30. (a) 31. (c) 32. (b) 33. (c) 34. (a) 35. (c) 36. (c) 37. (c) 38. (d) 39. (c)
 40. (a) 41. (b) 42. (c) 43. (c) 44. (a) 45. (a) 46. (b) 47. (c) 48. (c) 49. (c)
 50. (a)

Solutions with Necessary Explanation

1. $468 \div 39 = 12$

Therefore, 11 numbers are available and the number of pair of numbers having HCF = 39 is

given by $\frac{11+1}{2} = 6$

2. (i) The least number which is divisible by 6, 8 and 12 is the LCM of 6, 8 and 12 = 24.

The numbers greater than 500 and less than 700 which are divisible by 24 are

$$504, 528, 552, \dots, 696.$$

$$\begin{aligned}\text{The sum of the numbers} &= \frac{9}{2} (504 + 696) \\ &= 9 \times 600 = 5400\end{aligned}$$

- (ii) The number leaves remainder as 5 in each case, the required numbers are greater by 5 than the above numbers

$$\begin{aligned}\text{i.e., } &504 + 5, 528 + 5, \dots, 696 + 5 - 24 \\ &= 509, 533, \dots, 677.\end{aligned}$$

$$\begin{aligned}\text{The sum of the numbers} &= \frac{8}{2} (509 + 677) \\ &= 4 \times 1186 \\ &= 4744\end{aligned}$$

3. Since 1000 is not divisible by 15, it is not possible to divide 1000 into two parts such that their HCF may be 15. Hence the correct answer is (d).

4. Number of boys = 442; Number of girls = 374

As the number of boys in the class is same as that of girls.

$$\text{Strength of one class} = \text{HCF of } 442 \text{ and } 374 = 34$$

$$\therefore \text{Number of classes of boys} = 442 \div 34 = 13$$

$$\text{Number of classes of girls} = 374 \div 34 = 11$$

$$\therefore \text{Required number of classes} = 13 + 11 = 24$$

5. The greatest number of 4 digits = 9999

On dividing 9999 by 196, 3 will be left as remainder.

$$\therefore \text{The required greatest number} = 9996.$$

$$\text{Least number of 5 digits} = 10000$$

On dividing 10000 by 196, 4 will be left as remainder.

$$\therefore \text{The required 5 digit number} = 10000 - 4 + 196 = 10192$$

6. Let x and y be the two numbers and their HCF = 9; product = 4212.

$$\therefore x \times y = 9^2 \quad (\text{product in lower terms})$$

$$\text{i.e. } 4212 = 81 \quad (\text{product in lower terms})$$

$$\text{Product in lower terms} = 52$$